Author: Now let us return to the equality

$$\lim_{x\to a} f(x)=f(a) \quad (6)$$

You already know that there are situations when $\lim_{x\to a} f(x)$ exists but $f(a)$ does not exist and, vice versa, when $f(a)$ exists but $\lim_{x\to a} f(x)$ does not exist.

Finally, a situation is possible when both $\lim_{x\to a} f(x)$ and $f(a)$ exist, but their values are not equal. I will give you an example:

$f(x)=\begin{cases} x^2 & \text{if } x\neq 0 \\ 1 & \text{if } x=0 \end{cases}$ The graph of this function is shown in Fig. 28. It is easy to see that $f(0)=1$, while $\lim_{x\to 0} f(x)=0$.

You must be convinced by now that equality (6) is not always valid.

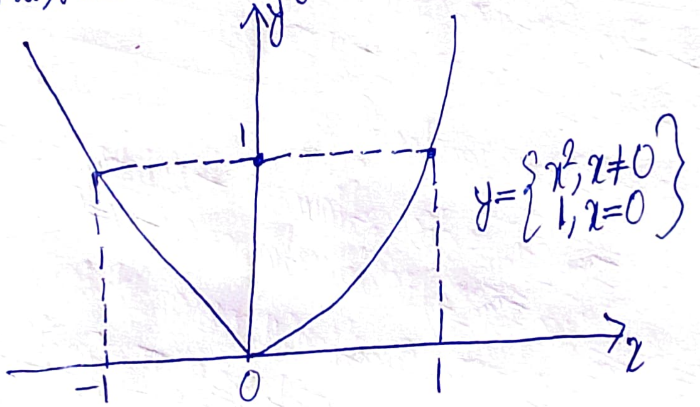

Fig. 28

Reader: But presumably, it is often true, isn't it?

Author: Yes, and if it is, the function $f(x)$ is said to be continuous at $x=a$.

Thus, we have arrived at a new important concept, namely, that of the continuity of a function at a point. Let us give the following

Definition:

A function $f(x)$ is said to be continuous at a point $x=a$ if (1) it is defined at $x=a$, (2) there is the limit of the function at $x=a$, (3) this limit equals the value of the function at $x=a$; or in other words, the function $f(x)$ is called continuous at a point a if $\boxed{\lim_{x\to a} f(x)=f(a)}$.

V. Imp Definition